# Mohammad Askari

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## EDUCATION

**UNIVERSITY OF CALIFORNIA, LOS ANGELES (UCLA)**

PhD Candidate in Civil Engineering — Expected Dec 2026

MSc Candidate in Statistics — Expected Dec 2026

## TECHNICAL EXPERIENCE

**UCLA** Los Angeles, CA — Sep 2021 - Present

Graduate Student Researcher

- Employed advanced statistical inference and machine learning methods to implement a Tweedie-XGBoost algorithm to estimate insurance losses for North Carolina Insurance Underwriting Association (NCIUA).
- Developed an end-to-end framework for building inventory generation through public data fusion and AI: using ViT-based and foundation models for image classification and segmentation followed by data imputation.
- Implemented a multi-objective optimization framework to calibrate simulated hurricane wind and flood hazards to ASCE-7 and FEMA targets.
- Designed and developed a parallelized, object-oriented Python package for regional hurricane risk analysis.
- Developed transformer-based NLP and ML techniques for PII removal and text classification of damage descriptions for NCIUA.
- Implemented GPU-based greedy search algorithm in OpenCL framework for ground motion selection. [GitHub]

## COURSE PROJECTS (SELECTED)

### Graduate-level

- Built an efficient ML pipeline that optimizes XGBoost hyperparameters via experimental design techniques, outperforming grid search in predicting wildfire damage to buildings. — Advisor: Professor Hongquan Xu
- Implemented Deep Q-learning with experience replay (DQN) and Deep Deterministic Policy Gradient (DDPG) reinforcement learning algorithms for Racing Car and Double Pendulium problems. — Advisor: Professor Jonathan Kao
- Evaluated performance of CRNN, LSTM, conditional GANs, and CNN models on time series data using TensorFlow, highlighting robust deep learning methods for signal processing. — Advisor: Professor Jonathan Kao [GitHub]
- A tutorial on the doubly robust targeted maximum likelihood (TMLE) estimation method and its implementation in R, merging causal inference with ML principles for improved estimation. — Advisor: Professor Chad Hazlett

## CERTIFICATIONS

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|---|---|---|
| Jul 2024 | TACC | AI Computational Academy & Hackathon |
| Jan 2024 | Coursera | AI for Good Specialization [Link] |
| Nov 2023 | NVIDIA | Building Transformer-Based NLP Applications [Link] |
| May 2023 | Coursera | Deep Learning Specialization [Link] |
| Feb 2021 | Coursera | Machine Learning [Link] |

## TECHNICAL SKILLS

Programming Languages: Python, R, C++

ML & DL Frameworks: PyTorch, JAX, TensorFlow, Keras, Scikit-learn, Transformers, LangChain, spaCy

Software Development Tools & External: Version Control (Git), GitHub, CI/CD, SQL, Bash, MATLAB, Docker

# PUBLICATIONS AND PRESENTATIONS

## Articles

[1] N. O. Tackie-Otoo, **M. Askari**, P. Hadinata, R. A. Davidson, E. Taciroglu, and G. Hardy, "Hurricane wind loss modeling using insurance claims data," *Natural Hazards*, vol. 122, no. 7, p. 300, 2026. doi: 10.1007/s11069-026-08059-z.

[2] **M. Askari**, M. H. Soleimani-Babakamali, C. Williams, R. Davidson, and E. Taciroglu, *CHEERSafe: A Python Library to Assess Hurricane Damage and Loss*, Manuscript in preparation, 2026.

[3] M. H. Soleimani-Babakamali, **M. Askari**, M. A. Heravi, R. Sisman, N. Attarchian, A. Askan, R. Soleimani, and E. Taciroglu, "Deep Ensemble Learning for Rapid Large-Scale Postearthquake Damage Assessment: Application to Satellite Images from the 2023 Türkiye Earthquakes," *ASCE OPEN: Multidisciplinary Journal of Civil Engineering*, vol. 3, no. 1, p. 04 025 003, 2025. doi: 10.1061/AOMJAH.AOENG-0043.

[4] S.-Z. Chen, X.-Y. Wang, X.-R. Ma, T. Wang, **M. Askari**, and E. Taciroglu, "Transforming Regression to Classification for Enhancing Extrapolation Capacity of Data-driven Structural Behavior Prediction," *Expert Systems with Applications*, vol. 277, p. 127 302, 2025, issn: 0957-4174. doi: https://doi.org/10.1016/j.eswa.2025.127302.

[5] R. Soleimani-Babakamali, M. H. Soleimani-Babakamali, M. A. Heravi, **M. Askari**, O. Avci, and E. Taciroglu, "Transferring Damage Detection Knowledge Across Rotating Machines and Framed Structures: Harnessing Domain Adaptation and Contrastive Learning," *Mechanical Systems and Signal Processing*, vol. 221, p. 111 743, 2024. doi: https://doi.org/10.1016/j.ymssp.2024.111743.

[6] M. Shokrabadi, Y. Bozorgnia, H. V. Burton, J. W. Baker, and **M. Askari**, "An Efficient Computational Platform for Selecting and Scaling Ground Motion Records while Considering Multiple Target Spectra," University of California, Los Angeles, Tech. Rep., 2022. doi: https://doi.org/10.34948/N3K01N.

[7] **M. Askari** and M. Mahsuli, "Modeling of Epistemic Uncertainties in Seismic Hazard Analysis Based on Reliability Methods," Center for Infrastructure Sustainability and Resilience Research, Sharif University of Tech., Tech. Rep., 2020.

## Presentations

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| May 2025 | EMI Conference | CHEERSafe: A Modular Object-Oriented Framework for Regional Hurricane Risk Assessment |
| May 2024 | PMC/EMI Conference | An End-To-End Framework for Building Inventory Generation Through Public Data Fusion and Artificial Intelligence |
| Feb 2024 | NHERI Computational Symposium | Deep Ensemble Learning for Rapid Large-Scale Post-Earthquake Damage Assessment—Application to 2023 Turkiye Earthquakes Satellite Images (*poster*) |
| Feb 2022 | ASCE Lifelines Conference | An Efficient Computational Platform for Selecting and Scaling Ground Motion Records while Considering Multiple Target Spectra (*poster*) |

# COMMUNITY ENGAGEMENT & VOLUNTEER EXPERIENCE

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| Sep 2024-present | Elsevier Engineering Structures<br>ASCE Open<br>ASCE I3CE | Journal article reviewer related to ML and AI applications |
| Apr 2024 | LA Hacks | Expert judgement for Los Angeles AI Hackathon event |
| Feb 2024-present | NSF | Code and documentation management of CoPe-CHEER NSF project |
| Sep 2022-July 2023 | UCLA | Board member of the Iranian Graduate and Professional Association, IGPA |